Ruocheng Wang

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Education

Shanghai Jiao Tong University & Shanghai Innovation Institute Sep. 2025 – Present
Direct-entry Ph.D. program (joint training).
Advisor: Prof. Junchi Yan, School of Artificial Intelligence.

Shanghai Jiao Tong University Sep. 2019 – Jun. 2025
B.Sc. in Physics, Zhiyuan College.

Honors and Awards

- National Scholarship for Graduate Students 2026
- Rong Chang Science and Technology Innovation Scholarship 2024, 2025
- Shanghai Outstanding Graduate 2025
- National Scholarship; A-Class Undergraduate Scholarship 2023
- Bronze Medal, Chinese Physics Olympiad (CPhO) 2019
- Jiangsu Provincial First Prize, Chinese Mathematical Olympiad (CMO) 2019

Publications

* Equal contribution. † Corresponding author.

Quantum Neural Operators with Implicit Quadratic Frame and Expressivity Advantages
Ruocheng Wang* (co-first author), Xiaoqiu Zhong*, Zhuo Xia, Junchi Yan†.
Nature Machine Intelligence

CoEvol-NO: State and Coordinate Co-Evolution with an Error-Driven Predictor-Corrector Paradigm for Neural Operator Transformer
Jianqiao Zeng, **Ruocheng Wang** (second author), Yanzhi Liu, Hao Xiong, Junchi Yan†.
ICML 2026 (Oral)

QuanONet: Quantum Neural Operator with Application to Differential Equation
Ruocheng Wang (first author), Zhuo Xia, Ge Yan, Junchi Yan†.
ICML 2025

Rethinking Parity Check Enhanced Symmetry-Preserving Ansatz
Ge Yan, Mengfei Ran, **Ruocheng Wang**, Kaiseng Pan, Junchi Yan†.
NeurIPS 2024

Non-Equilibrium and Self-Organization Evolution in Hot-Spot Ignition Processes
X.-Y. Fu*, Z.-Y. Guo*, Q.-H. Wang*, **R.-C. Wang*** (co-first author), D. Wu, J. Zhang†.
Physics of Plasmas

HomeGuard: VLM-based Embodied Safeguard for Identifying Contextual Risk in Household Task
Xiaoya Lu, Yijin Zhou, Zeren Chen, **Ruocheng Wang**, Bingrui Sima, Enshen Zhou, Lu Sheng, Dongrui Liu, Jing Shao.
ECCV 2026

AI-Empowered Quantum Circuit Synthesis and Compilation Optimization: Overview and Prospects
Ge Yan, Wenjie Wu, Yuheng Chen, Kaisen Pan, Xudong Lu, Zixiang Zhou, Yuhan Wang, **Ruocheng Wang**, Junchi Yan†.
TPAMI 2026

Research Projects

Recurrent RG Net: Renormalization Group Learning without Coupling Labels 2026 – Present

- Develop a framework for multiscale statistical learning without coupling labels, phase labels, or supervision from known critical points. Jointly learn a local coarse-graining operator C, low-dimensional theory coordinates Q, and a shared renormalization group map R from configuration ensembles.
- Design a joint training framework that combines consistency across ensemble subsets, data geometry across all scales, and one-step and multistep closed-loop constraints to learn finite-scale statistics and their latent-space dynamics.
- Completed prototype experiments with one- and two-dimensional latent coordinates on the one-dimensional transverse-field Ising model, qualitatively observing candidate stable, unstable, and saddle fixed points. Ongoing work investigates two-dimensional Ising systems and the reliable identification of critical behavior.

CoEvol-NO: A Neural Operator Transformer with State and Coordinate Co-Evolution Jan. 2025 – Jun. 2026

- Proposed an architecture that jointly evolves states and coordinates to address the trade-off between preserving geometric detail and retaining memory of long-term physical evolution in neural operators.
- Introduced a classical predictor-corrector paradigm that combines explicit target prediction with error-driven exact gradient correction, enabling the joint evolution of latent variables and mesh sequences.
- Proved that the architecture has strictly linear $\mathcal{O}(N)$ time complexity and that its exact gradient correction mechanism overcomes the instability of conventional residual updates under first-order approximation.
- Achieved state-of-the-art performance on industrial-scale fluid simulation tasks, including Shape-Net Car and AirfRANS, as well as standard benchmarks such as Navier–Stokes.

QuanONet: Quantum Neural Operators with Theoretical Expressivity Advantages Dec. 2024 – Apr. 2026

- Proposed QuanONet, the first quantum neural operator framework for the NISQ era, representing continuous nonlinear mappings as the Frobenius inner product between branch-modulated observables and a trunk density matrix.
- Extended the operator universal approximation theorem to the quantum setting and proved that the density matrix implicitly constructs a quadratic feature frame, surpassing the linear capacity limit of classical models to achieve $\mathcal{O}(p^2)$ expressive capacity.
- Introduced a trainable-frequency (TF) embedding strategy that adaptively aligns with the dominant eigenfeatures of the output covariance operator, substantially improving the capture of broadband dynamics.
- Demonstrated strong optimization robustness in high-dimensional scaling regimes and validated the framework on real IBM superconducting quantum processors.